

Algebra II with Trigonometry

Chapter 5.6

Olympiad Problems

Saved tutoring response with MathJax rendering. Original PDF file:

[4401272_alg-2trig-h-chp-5-6-note-docx.pdf](#)

Problems

1. Given triangle ABC where $AB = AC = 13$ and $BC = 10$, find $\cos A$ without evaluating any cosines or square roots explicitly.
2. In triangle XYZ, the three sides are consecutive integers and the largest angle is exactly twice the smallest angle. Determine the possible values of the sides.
3. A cyclist travels 8 km due north, then 6 km due east, and finally 5 km in a straight line back towards the starting point. Find the angle between the first and last leg, expressed exactly in degrees.
4. Consider a triangle with side lengths a , b , c . If the area of the triangle is exactly one-fourth its perimeter, and $a = 14$, $b = 13$, find c .
5. Two triangles share two sides of lengths 7 and 15, but their included angles differ by 60° . What is the difference between the squares of the third side of each triangle?
6. The points A, B, and C lie in the plane where $AB = 5$, $BC = 6$, and $CA = 7$. Show that $\cos \angle A + \cos \angle B + \cos \angle C$ can be written as a rational number, and find its value.